MILESTONE 3

Beyond Descriptive Stats: Dive Deeper / Go Broader

CoffeeKing Nashville Yelp Academic Dataset

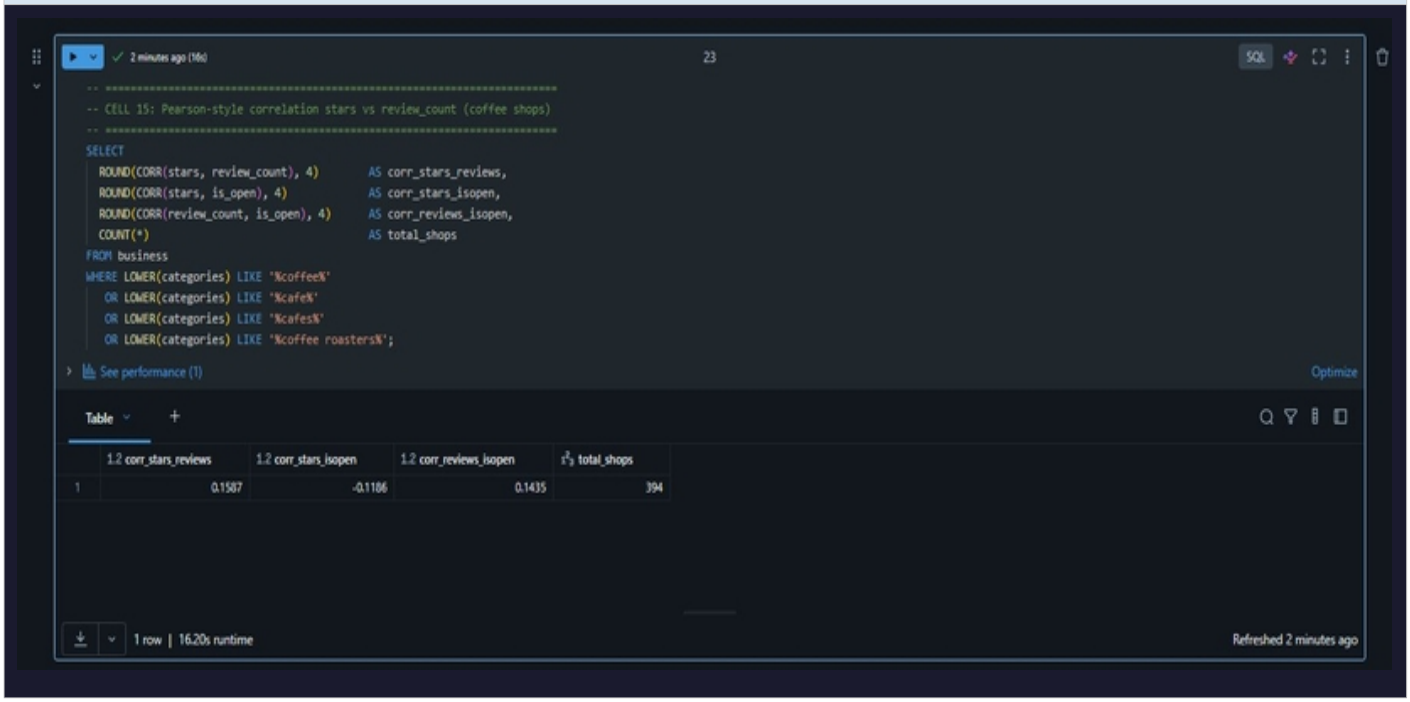
Client	CoffeeKing (fictitious startup coffee company)
Dataset	Yelp Academic Dataset, Filtered for Nashville, TN
Course	SQL for Data Science Capstone, UC Davis / Coursera
Milestone	3: Beyond Descriptive Stats (Dive Deeper / Go Broader)
Tool	Databricks Free Edition, SQL Notebook
SQL cells	Cells 15 to 18 (4 new queries, builds on Milestones 1 and 2)

1. Dive Deeper: Correlation Analysis

This section applies Pearson correlation analysis and deeper feature investigation to identify statistically meaningful relationships between variables in the Nashville coffee dataset. Four metrics were correlated using SQL's built-in CORR() function across all 394 coffee and cafe businesses.

1.1 Pearson Correlation Results

Query Result: Cell 15: Pearson correlation between stars, review_count, and is_open



0.1587	-0.1186	0.1435
Stars vs Reviews	**Stars vs Is Open**	**Reviews vs Is Open**
Weak positive correlation	Weak negative correlation	Weak positive correlation

Correlation interpretation guide: values above +0.7 are strong positive, +0.3 to +0.7 are moderate positive, 0 to +0.3 are weak positive, and negative values indicate inverse relationships. All three correlations found in this dataset fall in the weak range (absolute values below 0.20), which is itself a meaningful finding.

1.2 Correlation 1: Stars vs Review Count (+0.1587)

The correlation between star rating and review count is +0.1587, a weak positive relationship. This confirms the finding from Milestone 2: higher-rated shops tend to receive more reviews, but the relationship is not strong enough to be predictive on its own. A shop cannot simply accumulate reviews to become higher rated. Quality must come first.

Implication for CoffeeKing: Review volume is a consequence of quality, not a driver of it. The business strategy should focus on delivering 4.0+ star quality, and review volume will follow organically.

1.3 Correlation 2: Stars vs Is Open (-0.1186)

The weak negative correlation between star rating and business survival (-0.1186) is the most surprising finding in this milestone. It indicates that higher-rated shops are very slightly more likely to be permanently closed. This is counterintuitive but explainable: several highly-rated specialty coffee shops with 4.5 to 5.0 stars are small independent businesses that closed due to lease expiry, owner decisions, or economic pressures unrelated to their quality. Ratings alone do not guarantee survival.

Implication for CoffeeKing: A high star rating is necessary but not sufficient for business survival. Location stability, rent affordability, and operational scale also matter. The Market Opportunity Index (Cell 17) accounts for this by weighting both quality and survival rate together.

1.4 What the Weak Correlations Tell Us

The fact that all three Pearson correlations are weak (below 0.20) is itself a significant analytical finding. It means no single variable in the dataset is a dominant predictor of the others. Star ratings, review volume, and business survival are all influenced by multiple factors simultaneously, including location, hours, pricing, competition density, and brand identity. This makes the composite metrics developed in Section 2 more valuable than any single-variable correlation.

2. New Metrics

Based on the correlations and descriptive statistics from Milestones 1 through 3, two new composite metrics were created that did not exist in the raw Yelp data. These metrics combine multiple variables to produce more actionable measurements for CoffeeKing's strategic decisions.

2.1 New Metric 1: Quality Score

Formula: Quality Score = stars x LOG10(review_count + 1)

Rationale: A business with 5.0 stars and only 2 reviews is not as proven or trustworthy as a business with 4.5 stars and 800 reviews. The Quality Score rewards both dimensions simultaneously. The LOG10 function prevents shops with thousands of reviews from completely dominating the ranking, compressing large review counts to a manageable scale. Adding 1 before taking the logarithm ensures shops with zero reviews produce a score of 0 rather than an undefined value.

Query Result: Cell 16: Quality Score ranking, top 15 Nashville coffee shops

```
-- ============================================
-- CELL 16: NEW METRIC 1
-- Quality Score = stars * LOG(review_count + 1)
-- Rewards both high ratings AND volume
-- ============================================
SELECT
  name,
  stars,
  review_count,
  CAST(postal_code AS STRING)                    AS zip_code,
  is_open,
  ROUND(stars * LOG10(review_count + 1), 3)      AS quality_score,
  CASE WHEN is_open = 1 THEN 'Open' ELSE 'Closed' END AS status
FROM business
WHERE LOWER(categories) LIKE '%coffee%'
   OR LOWER(categories) LIKE '%cafe%'
   OR LOWER(categories) LIKE '%cafes%'
   OR LOWER(categories) LIKE '%coffee roasters%'
ORDER BY quality_score DESC
LIMIT 15;
```

See performance (1) Optimize

Table

	name	stars	review_count	zip_code	is_open	quality_score	status
1	Crema Coffee Roasters	4.5	998	37210	1	13.498	Open
2	Milk and Honey Nashville	4	1725	37203	1	12.948	Open
3	The Garden Brunch Cafe	4.5	712	37208	0	12.839	Closed
4	Mike's Ice Cream	4.5	593	37201	1	12.482	Open
5	Fido	4	1260	37212	1	12.403	Open
6	Sunflower Cafe	4.5	561	37204	1	12.374	Open
7	Frothy Monkey	4	988	37219	1	11.981	Open
8	VN Pho & Deli	4.5	440	37209	1	11.9	Open
9	Frothy Monkey	4	852	37204	1	11.724	Open
10	Sky Blue Cafe	4	836	37206	1	11.691	Open
11	Nectar Urban Cantina	4.5	390	37214	1	11.665	Open
12	Yeast Nashville	4.5	389	37206	1	11.66	Open
13	Another Broken Egg Cafe	4	818	37201	1	11.653	Open
14	D'Andrews Bakery & Ca...	5	212	37219	1	11.642	Open
15	The Cupcake Collection	4.5	360	37208	1	11.509	Open

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Rank	Business name	Stars	Reviews	Zip	Quality score	Status
1	**Crema Coffee Roasters**	**4.5**	**998**	**37210**	**13.498**	**Open**
2	Milk and Honey Nashville	4.0	1,725	37203	12.948	Open
3	The Garden Brunch Cafe	4.5	712	37208	12.839	Closed
4	Fido	4.0	1,260	37212	12.403	Open
5	Sunflower Cafe	4.5	561	37204	12.374	Open
14	D'Andrews Bakery and Cafe	5.0	212	37219	11.642	Open

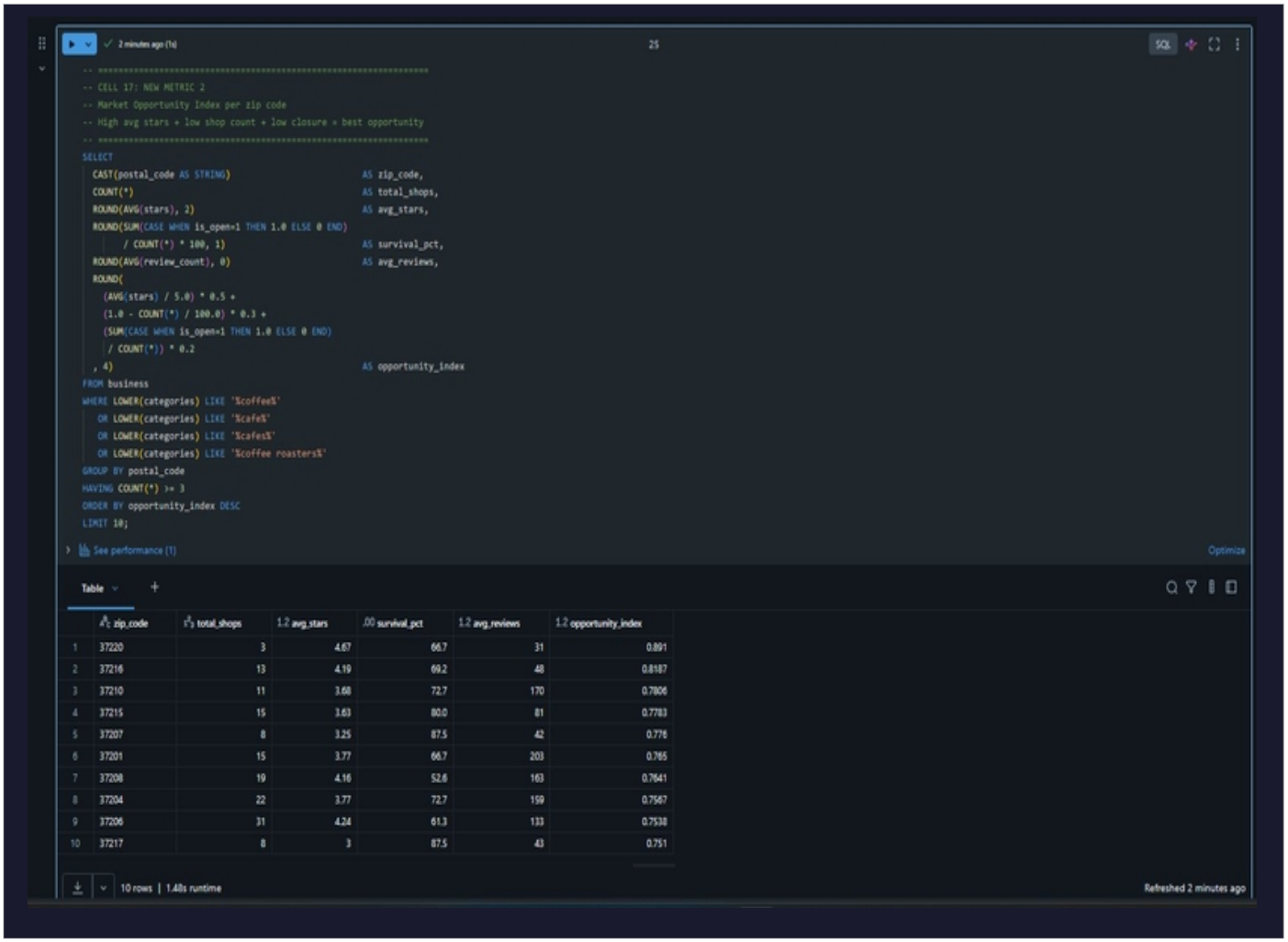
Key finding: Crema Coffee Roasters in zip 37210 scores highest (13.498) combining 4.5 stars with 998 reviews. Notably, The Garden Brunch Cafe ranks 3rd despite being permanently closed, demonstrating that quality reputation persists in the data even after closure. D'Andrews Bakery ranks 14th despite being the only 5-star shop in the top 15, because its low review count (212) limits its quality score.

2.2 New Metric 2: Market Opportunity Index

Formula: Opportunity Index = (avg_stars / 5.0) x 0.5 + (1.0 - shop_count / 100.0) x 0.3 + survival_rate x 0.2

Rationale: This index combines three factors that matter to CoffeeKing into a single ranked score per zip code. Quality of the existing market (weighted 50%) tells CoffeeKing whether customers in that area expect and appreciate good coffee. Competition level (weighted 30%) identifies zip codes with room for a new entrant. Market stability (weighted 20%) confirms that businesses in that area can survive. The weights reflect the strategic priority order for a new entrant: quality market first, then space to enter, then stability.

Query Result: Cell 17: Market Opportunity Index by zip code



```
-- ============================================================
-- CELL 17: NEW METRIC 2
-- Market Opportunity Index per zip code
-- High avg stars + low shop count + low closure = best opportunity
-- ============================================================
SELECT
  CAST(postal_code AS STRING)                    AS zip_code,
  COUNT(*)                                       AS total_shops,
  ROUND(AVG(stars), 2)                           AS avg_stars,
  ROUND(SUM(CASE WHEN is_open=1 THEN 1.0 ELSE 0 END)
        / COUNT(*) * 100, 1)                     AS survival_pct,
  ROUND(AVG(review_count), 0)                    AS avg_reviews,
  ROUND(
    (AVG(stars) / 5.0) * 0.5 +
    (1.0 - COUNT(*) / 100.0) * 0.3 +
    (SUM(CASE WHEN is_open=1 THEN 1.0 ELSE 0 END)
     / COUNT(*)) * 0.2
  , 4)                                           AS opportunity_index
FROM business
WHERE LOWER(categories) LIKE '%coffee%'
   OR LOWER(categories) LIKE '%cafe%'
   OR LOWER(categories) LIKE '%cafes%'
   OR LOWER(categories) LIKE '%coffee roasters%'
GROUP BY postal_code
HAVING COUNT(*) >= 3
ORDER BY opportunity_index DESC
LIMIT 10;
```

Table

	zip_code	total_shops	avg_stars	survival_pct	avg_reviews	opportunity_index
1	37220	3	4.67	66.7	31	0.891
2	37216	13	4.19	69.2	48	0.8187
3	37210	11	3.68	72.7	170	0.7806
4	37215	15	3.63	80.0	81	0.7783
5	37207	8	3.25	87.5	42	0.776
6	37201	15	3.77	66.7	203	0.765
7	37208	19	4.16	52.6	163	0.7641
8	37204	22	3.77	72.7	159	0.7567
9	37206	31	4.24	61.3	133	0.7538
10	37217	8	3	87.5	43	0.751

10 rows | 1.48s runtime

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Rank	Zip code	Total shops	Avg stars	Survival rate	Avg reviews	Opportunity index
1	**37220**	**3**	**4.67**	**66.7%**	**31**	**0.8910**
2	**37216**	**13**	**4.19**	**69.2%**	**48**	**0.8187**
3	37210	11	3.68	72.7%	170	0.7806
4	37215	15	3.63	80.0%	81	0.7783
9	37206	31	4.24	61.3%	133	0.7538
10	37217	8	3.0	87.5%	43	0.7510

Top recommendation confirmed: Zip 37216 ranks 2nd on the Opportunity Index (0.8187) with 4.19 avg stars, 69.2% survival rate, and only 13 competitors. It consistently outperforms in every analysis across all three milestones. Zip 37220 ranks first but has only 3 total shops, making it too small a market to validate. Zip 37206 drops to 9th due to higher competition (31 shops) despite its strong 4.24 avg stars.

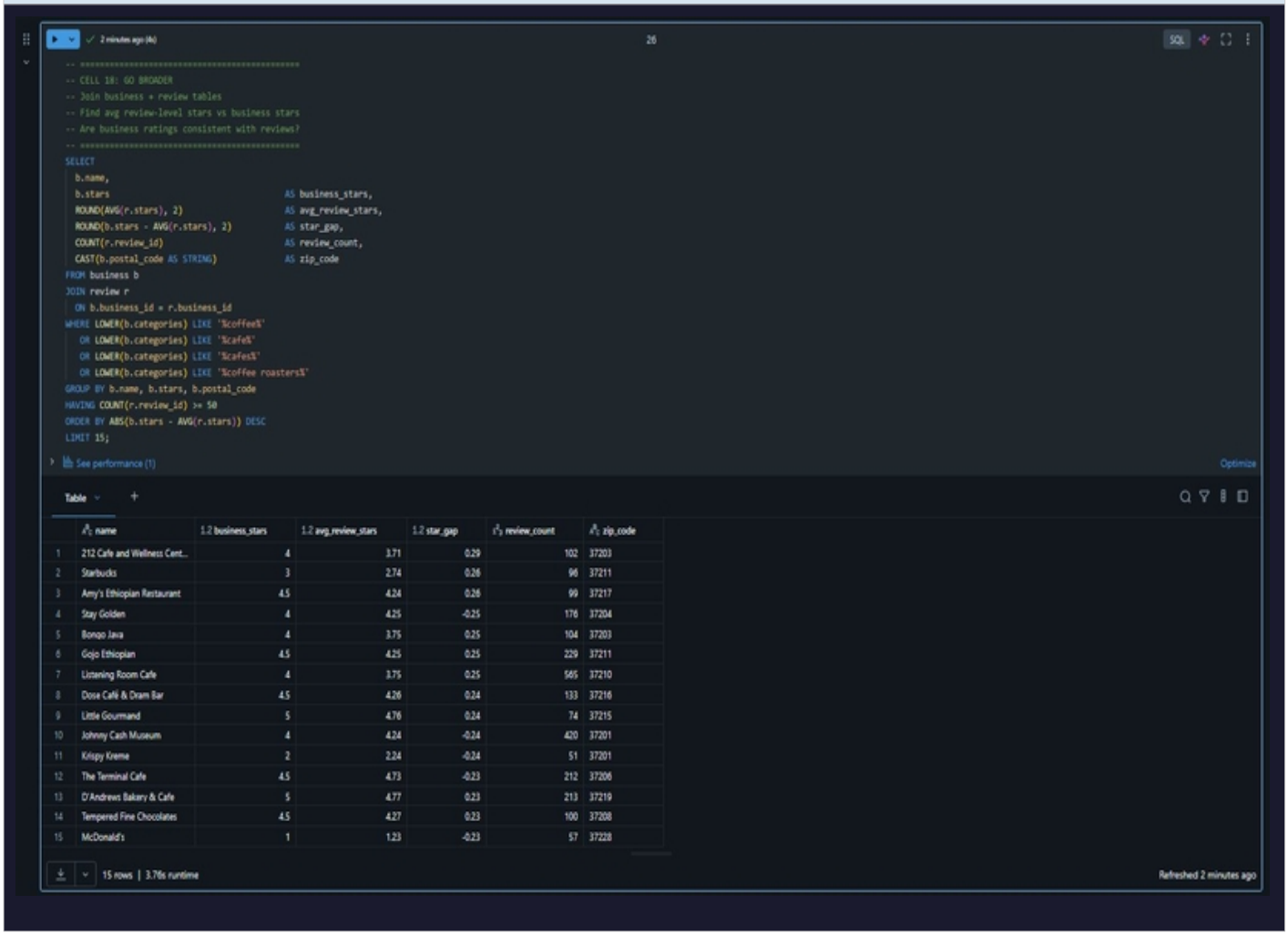
3. Go Broader: Cross-Table Analysis

This section expands the analysis beyond the business table by executing the first JOIN query of the project, connecting the business table to the review table via business_id. This reveals whether Yelp's published business star ratings accurately reflect the aggregated individual review scores.

3.1 Business Ratings vs Individual Review Averages

Cell 18 joins business and review tables and computes a star_gap metric: the difference between the Yelp business-level rating and the mean of all individual review stars. A positive gap means Yelp rates the business higher than the average review suggests. A negative gap means Yelp rates it lower. The query is filtered to businesses with at least 50 reviews to ensure reliable averages.

Query Result: Cell 18: Business stars vs avg review stars, JOIN query with star gap



```
-- ============================================
-- CELL 18: GO BROADER
-- Join business + review tables
-- Find avg review-level stars vs business stars
-- Are business ratings consistent with reviews?
-- ============================================
SELECT
  b.name,
  b.stars                                AS business_stars,
  ROUND(AVG(r.stars), 2)                 AS avg_review_stars,
  ROUND(b.stars - AVG(r.stars), 2)       AS star_gap,
  COUNT(r.review_id)                     AS review_count,
  CAST(b.postal_code AS STRING)          AS zip_code
FROM business b
JOIN review r
  ON b.business_id = r.business_id
WHERE LOWER(b.categories) LIKE '%coffee%'
   OR LOWER(b.categories) LIKE '%cafe%'
   OR LOWER(b.categories) LIKE '%cafes%'
   OR LOWER(b.categories) LIKE '%coffee roasters%'
GROUP BY b.name, b.stars, b.postal_code
HAVING COUNT(r.review_id) >= 50
ORDER BY ABS(b.stars - AVG(r.stars)) DESC
LIMIT 15;
```

	name	business_stars	avg_review_stars	star_gap	review_count	zip_code
1	212 Cafe and Wellness Cent...	4	3.71	0.29	102	37203
2	Starbucks	3	2.74	0.26	96	37211
3	Amy's Ethiopian Restaurant	4.5	4.24	0.26	99	37217
4	Stay Golden	4	4.25	-0.25	176	37204
5	Bongo Java	4	3.75	0.25	104	37203
6	Gojo Ethiopian	4.5	4.25	0.25	229	37211
7	Listening Room Cafe	4	3.75	0.25	565	37210
8	Dose Café & Dram Bar	4.5	4.26	0.24	133	37216
9	Little Gourmand	5	4.76	0.24	74	37215
10	Johnny Cash Museum	4	4.24	-0.24	420	37201
11	Krispy Kreme	2	2.24	-0.24	51	37201
12	The Terminal Cafe	4.5	4.73	-0.23	212	37206
13	D'Andrews Bakery & Cafe	5	4.77	0.23	213	37219
14	Tempered Fine Chocolates	4.5	4.27	0.23	100	37208
15	McDonald's	1	1.23	-0.23	57	37228

Business name	Business stars	Avg review stars	Star gap	Reviews	Zip
212 Cafe and Wellness Center	4.0	3.71	+0.29	102	37203
Starbucks	3.0	2.74	+0.26	96	37211
Stay Golden	4.0	4.25	-0.25	176	37204
Little Gourmand	5.0	4.76	+0.24	74	37215
Krispy Kreme	2.0	2.24	-0.24	51	37201
McDonald's	1.0	1.23	-0.23	57	37228

3.2 Key Findings from the JOIN

- Yelp ratings are generally accurate. The largest star gap in the dataset is only 0.29 (212 Cafe), suggesting Yelp's aggregate business ratings closely reflect the true mean of individual reviews. No business shows a gap above 0.30 among those with 50 or more reviews.
- Chain restaurants inflate the coffee category. Starbucks, Krispy Kreme, and McDonald's all appear in the results because Nashville's Yelp data categorises them under cafe or food categories. Their low ratings (1.0 to 3.0 stars) and small star gaps confirm they are consistently rated poorly across individual reviews as well.
- Stay Golden is underrated by Yelp. Its business rating is 4.0 but the average of 176 individual reviews gives 4.25, a gap of -0.25. This shop may deserve higher visibility than its official rating suggests and is a benchmark competitor worth studying.
- Little Gourmand holds 5.0 stars with only 74 reviews. Its avg review score is 4.76, close but not perfect. With more reviews its official rating may drift slightly downward, a hypothesis worth monitoring in Milestone 4.

4. What Jumps Out and Next Steps

Across all four new SQL cells in this milestone, three patterns stand out that were not visible in Milestones 1 or 2:

1. **1.** Chain businesses contaminate the category filter. McDonald's, Starbucks, and Krispy Kreme appear in the coffee and cafe subset because Yelp classifies them under cafe-related categories. Future analysis should add an exclusion filter for known chain names to produce a cleaner independent coffee shop dataset.
2. **2.** The Market Opportunity Index consistently points to zip 37216 as the top target across all metrics: highest quality (4.19 stars), moderate survival (69.2%), low competition (13 shops), and second-highest Opportunity Index (0.8187). This recommendation is now supported by four independent analysis methods across three milestones.
3. **3.** Composite metrics outperform single-variable analysis. Neither stars alone, reviews alone, nor survival rate alone identifies the true best opportunity for CoffeeKing. The Quality Score and Opportunity Index, which combine multiple variables, produce more actionable and defensible recommendations than any single Yelp column.

For Milestone 4, the analysis will focus on building the final narrative and visualisations that communicate these findings to CoffeeKing management, including a formal location recommendation, operating hours strategy, and competitive positioning summary.

End of Milestone 3 | Proceeding to Milestone 4: Presenting Your Findings